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Asymptotic limits and wrinkling patterns in a pressurised shallow spherical cap

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ABSTRACT

The asymmetric bifurcation problem for a shallow spherical cap is examined. The applied pressure can act either external or internal to the cap and both cases are treated here. Assuming a non-linear axisymmetric basic state, the linearised bifurcation equations for the pressurised shell are investigated in the limit when the thickness of the cap is much less than the maximum rise of the shell mid-surface. Within this regime the wrinkling patterns in both cases are confined to a narrow zone near the edge of the shell, making it possible to solve asymptotically the corresponding equations and derive analytical predictions for both the critical pressure and the corresponding number of wrinkles. Some comparisons with direct numerical simulations are included as well.

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1. Introduction

Interest in the buckling of an externally pressurised shallow spherical cap was originally motivated by research into the behaviour of a complete spherical shell subjected to the same type of loading. Experimental evidence indicates that the full shell deforms and is susceptible to a mode characterised by a large number of adjacent circular dimples that cover the entire surface of the sphere [1], thereby giving an appearance not dissimilar to that of a golf ball. Since each of the buckles subtends only a small solid angle at the centre of the sphere, it was a small step to conclude that an understanding of the shallow cap problem with a clamped edge might shed light on the behaviour of spherical shells. Of course, this analogy between the two configurations is only partially valid, but the buckling and post-buckling behaviours of spherical caps have remained a classic topic and received a considerable amount of attention in the literature (e.g., see the recent survey [2] and the references therein). Much of this interest can be traced back to the widespread use of spherical structural elements, with geometries ranging from hemispherical to reasonably shallow. For instance, shallow and deep spherical caps are found extensively in powered submersibles [3], while in the nuclear power industry they are used in pressure safety devices [4,5]. Very recently shallow spherical caps have found increasing applicability in the design of transducers (e.g., [6]).

When subjected to a uniform *external* pressure field (i.e., directed towards the centre of curvature of the cap), a shallow spherical cap can display two distinct modes of deformation, as illustrated in the schematic Fig. 1. So-called snap-through buckling, as illustrated in the left window, is characterised by a sudden reversal of curvature over a small region near the apex and, typically, involves large displacements. This type of instability is the more studied case because the deformation experienced by the shell is axisymmetric and the governing equations are ordinary rather than partial differential equations. Detailed historical surveys regarding the experimental development of this problem can be found in the books by Bushnell [1] or Singer et al. [7] while theoretical aspects are reviewed by Shilkrut [8], with additional excellent accounts being found in [9,10].

Snap-through buckling is usually associated with the presence of a limit point on the load–(apex) displacement curve for the cap, although the instability is sometimes triggered before that point is reached. Loss of stability through this route only occurs over a relatively narrow range of values of the height-to-thickness ratio of the cap. If this ratio is very small then no instability occurs, whereas for larger values the shell tends to experience *azimuthal buckling* or *wrinkling*, as illustrated in the right window of Fig. 1. If the aforementioned ratio is large ($\simeq 15$ or greater) then the wrinkles tend to concentrate near the shell edge. The governing equations describing this type of bifurcations can still be reduced to ordinary differential equations due to the regular nature of the

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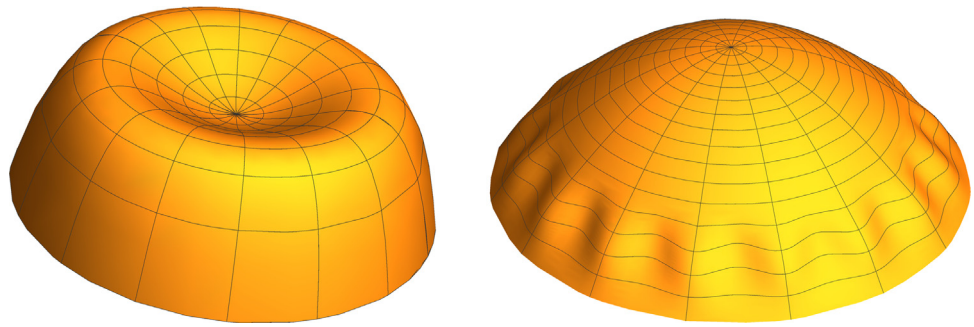

Fig. 1. An illustration of the two principal modes of instability experienced by an externally pressurised spherical shell: *snap-through buckling* (left), and *asymmetric wrinkling* (right); the deformation modes are not to scale.

wrinkling pattern, but they are of a much higher order than those for snap-through buckling.

Weinitschke [11] was the first to systematically calculate the critical pressures that trigger asymmetric buckling in shallow spherical caps. He adopted power series representations for both the non-linear basic state and the bifurcation equations, but the values predicted by his approach turned out to be quite low compared with those seen in practice. Huang [12] revisited the same problem but replaced the power series with a finite-differences numerical scheme that allowed more accurate results to be derived. Archer and Famili [13] confirmed the improved accuracy of Huang's results by using an alternative approach to buckling based on dynamical bifurcation equations. This involved examining the asymmetric frequency spectrum corresponding to small free vibrations of the shell in the neighbourhood of the non-linear axisymmetric equilibrium state. The buckling mode and corresponding critical load were deduced by observing the vanishing of a particular frequency. These authors also developed a load-perturbation approach [14] to compute numerically the incipient stage for the asymmetric post-buckling response. Fitch [15] modified Huang's numerical solution to account for the problem when a vertical load concentrated at the apex of the cap is applied, and concluded that the bifurcation pattern is characterised by between three and five circumferential waves. This continues to be the case even for large height-to-thickness ratios but in this limit the deformation is confined near the apex rather than the edge of the shell. He also used a Lyapunov–Schmidt scheme to compute the initial post-buckling behaviour and established that the bifurcation is super-critical, with the consequence that the cap under a concentrated load is an imperfection-insensitive structure.

The related situation when the shallow spherical cap is subjected to a pressure acting in the opposite direction, that is from the inside of the cap, has received considerably less attention. This configuration, which we shall henceforth refer to as the *internally pressurised shell*, was studied numerically by Shilkrut [8,16] who seems to have been the first to point out that these structures undergo asymmetric buckling that is sometimes localised near the rim of the shell. Since most of the numerical information in his book [8] is presented in terms of the so-called 'deformation maps' it is somewhat arguable as to the extent to which the analogy between the externally and internally pressurised shallow spherical caps can be pursued. One aim of our present study is to throw some more light on the possible connections between the two problems by conducting an in-depth numerical exploration, and thereby establish the asymptotic structure of the corresponding bifurcation equations when the height-to-thickness ratio is large.

Additional motivation is provided by some recent related work [18,19] in which we explored the edge-wrinkling of pressurised thin elastic plates. As a flat circular plate may be thought of in terms of a degenerate spherical cap, it is of interest to gain some further understanding of the similarities and differences between the two configurations in the context of asymmetric bifurcations. The transition from shell to plate seems to have received scant attention in the literature although a noteworthy exception is the intriguing work [20]. In this study a particular type of spherical cap was pulled at the rim (see [21] for details) but deformations were restricted to axisymmetric forms.

To keep the paper reasonably self-contained, we commence our study by reviewing the key equations of interest and identify some suitable dimensionless parameters. As shown in Sections 2 and 3 the externally and internally pressurised shallow shell problems can be described by almost the same differential equations, with the only discrepancy arising from a sign change that indicates the direction of the applied pressure (see comments immediately after Eq. (2.5)). The boundary conditions for the two problems are somewhat different and these are spelled out in detail in those introductory sections. We proceed in Section 4 to conduct a numerical investigation of the dependence of the lowest critical wrinkling pressure on various quantities of interest. Further numerical work provides evidence as to the mechanisms responsible for the localisation of the wrinkling patterns, and samples of representative eigenmodes obtained. Guided by the numerical evidence of Section 4, we then discuss the asymptotic structure of the externally pressurised shallow cap in Section 5 and compare with the results of the numerical simulations. A parallel asymptotic investigation is carried out in Section 6 for the internally pressurised shell. The paper concludes with a discussion of our main findings and some remarks on possible extensions.

2. The basic state

We consider a shallow spherical cap of uniform thickness $h>0$, whose middle surface can be represented by the elliptic paraboloid $z=H\left[1-(r/a)^2\right]$, where H is the rise of the middle surface at the centre, and a denotes the base radius. The geometry of the configuration is sketched in Fig. 2; with our chosen dimensions the curvature radius of the shell is $R\equiv a^2/(2H)$ and the cap is shallow so long as $0<h/R\ll 1$. The linearly isotropic elastic material of the spherical cap is described by Young's modulus E and the Poisson ratio ν. Our two problems of interest are distinguished by the direction of application of the uniform pressure. In the first case, the shell experiences an external pressure P normal to its surface (see Fig. 3(a)), and we shall refer to this problem as $\mathcal{C}_{\rm sph}^{(-)}$. The second situation, as depicted in Fig. 3(b) and denoted by $\mathcal{C}_{\rm sph}^{(+)}$, is the far less studied problem of an outwardly pressurised spherical cap. Finally, it will be further assumed that suitable edge restraints are applied, and this will be made more precise shortly.

The starting point for setting up the relevant bifurcation problem is the well-known Donnell–Mushtari–Vlasov (DMV) shallow

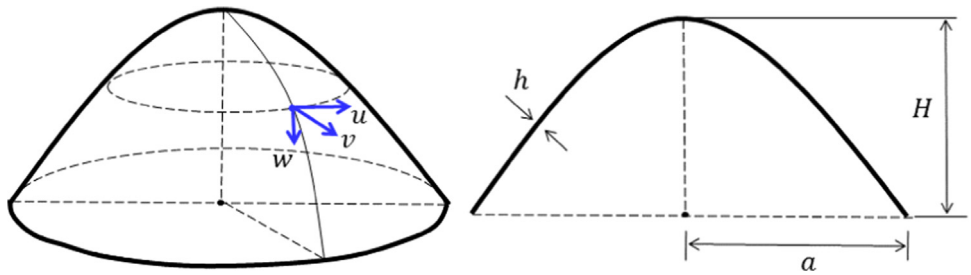

Fig. 2. The geometry of the shallow shell together with the directions of the components of the associated displacements.

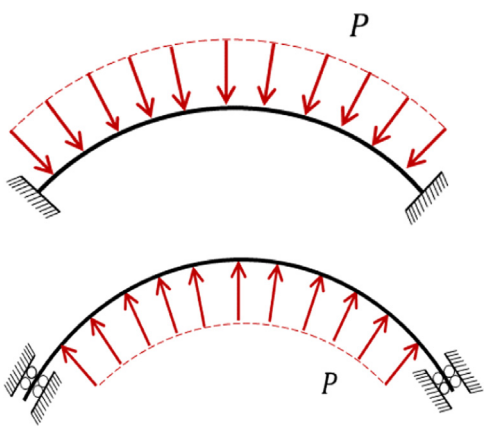

Fig. 3. The two configurations for the pressurised spherical cap: the problems $\mathcal{C}_{\rm sph}^{(-)}$ (top) and $\mathcal{C}_{\rm sph}^{(+)}$ (bottom). The externally pressurised shell is assumed to be clamped along its outer perimeter, while the other case involves a clamped, but movable edge (further explanation in the text).

shell equations (e.g., see [10,28])

$$D\nabla^4 w-\frac{1}{R}\nabla^2 F-[F,w]=P, \tag{2.1a}$$

$$\frac{1}{Eh}\nabla^4 F+\frac{1}{R}\nabla^2 w+\frac{1}{2}[w,w]=0, \tag{2.1b}$$

where $D\equiv Eh^3/12(1-\nu^2)$ is the bending rigidity, the bracket represents the Monge–Ampère operator defined by

$$[f,g]:=(\nabla^2 f)(\nabla^2 g)-(\nabla\otimes\nabla f):(\nabla\otimes\nabla g)$$

for any two smooth functions f and g, and the stress function F is related to the membrane tensor $\boldsymbol{N}$ according to

$$\boldsymbol{N}=(\nabla^2 F)\boldsymbol{I}_2-\nabla\otimes\nabla F, \tag{2.2}$$

with $\boldsymbol{I}_2$ denoting the two-dimensional identity tensor. We note in passing that the Föppl–von Kármán equations can be recovered formally from (2.1) by taking the limit $R\to\infty$.

Eqs. (2.1) are rescaled by introducing suitable dimensionless quantities defined by

$$\rho:=\frac{r}{a},\quad \mu:=2\left[3(1-\nu^2)\right]^{1/4}\left(\frac{H}{h}\right)^{1/2},\quad P_{\rm sph}:=32E\mu^{-2}\left(\frac{H}{a}\right)^4\left(\frac{h}{H}\right), \tag{2.3a}$$

$$\lambda:=\frac{P}{P_{\rm sph}},\quad \overline{w}:=\frac{\mu^2 w}{2H},\quad \overline{F}:=\frac{\mu^4 F}{4EH^2 h}; \tag{2.3b}$$

here $P_{\rm sph}$ is the classical buckling pressure of a complete spherical shell of the same radius of curvature and thickness. If for the moment we restrict our attention to axi-symmetric solutions, that is, $\overline{w}(\rho,\theta)\equiv\overline{w}(\rho)=:\mathring{w}(\rho)$ and $\overline{F}(\rho,\theta)\equiv\overline{F}(\rho)=:\mathring{F}(\rho)$, then it can be shown that (2.1) is easily expressed in the equivalent form

$$\mathcal{C}_{\rm sph}^{(\pm)}\ \mathcal{L}_0[\Theta]+\mu^2\Phi=\pm 2\mu^4\lambda\rho+\frac{\Theta\Phi}{\rho}, \tag{2.4a}$$

$$\mathcal{L}_0[\Phi]-\mu^2\Theta=-\frac{\Theta^2}{2\rho}, \tag{2.4b}$$

where we have introduced the new dependent variables

$$\Theta\equiv\Theta(\rho;\lambda,\mu):=-\frac{d\mathring{w}}{d\rho},$$
$$\Phi\equiv\Phi(\rho;\lambda,\mu):=\frac{d\mathring{F}}{d\rho}, \tag{2.5}$$

and the operator

$$\mathcal{L}_0\equiv\frac{d^2}{d\rho^2}+\frac{1}{\rho}\frac{d}{d\rho}-\frac{1}{\rho^2}.$$

Eqs. (2.4) describe both problems with $\lambda>0$; the only difference is that for the external (internal) problem $\mathcal{C}_{\rm sph}^{(-)}$ ($\mathcal{C}_{\rm sph}^{(+)}$) the minus (plus) sign needs to be used on the first term on the right-hand side of (2.4a). For an externally pressurised shell with a completely clamped rim the system (2.4) needs to be solved subject to

$$\mathcal{C}_{\rm sph}^{(-)}:\quad \begin{cases}\Theta(1)=0,\quad \dfrac{d\Phi}{d\rho}(1)-\nu\Phi(1)=0, & (2.6a)\\ \Theta(0)=0,\quad \Phi(0)=0; & (2.6b)\end{cases}$$

the two conditions in (2.6b) arise from the regularity requirements that together ensure the membrane forces remain bounded at the apex.

As already pointed out in Section 1, for an internally pressurised spherical cap the choice of boundary conditions is dictated by the necessity of having compressive circumferential stresses for the instability to occur. One such possibility corresponds to a clamped, but movable edge (i.e., an edge that can move freely in the radial direction but cannot rotate), and in this case the following set of mathematical constraints must be adopted

$$\mathcal{C}_{\rm sph}^{(+)}:\quad \begin{cases}\Theta(1)=0,\quad \Phi(1)=0, & (2.7a)\\ \Theta(0)=0,\quad \Phi(0)=0. & (2.7b)\end{cases}$$

The presence of both bending and membrane boundary layers in shallow shell equations similar to (2.4) subject to simply supported or clamped conditions was first established by Reissner [22,23], but no attempt was made to exploit this in a systematic manner.

3. The bifurcation boundary-value problem

Bifurcations from the radially symmetric solutions that satisfy the system (2.4) subject to either (2.6) or (2.7), as appropriate, are obtained by using the method of adjacent equilibrium. This involves considering perturbations of the form $\overline{w}(\rho,\theta)\to\mathring{w}(\rho)+\widehat{w}(\rho,\theta)$ and $\overline{F}(\rho,\theta)\to\mathring{F}(\rho)+\widehat{F}(\rho,\theta)$ that are then substituted in the non-dimensional version of (2.1), followed by

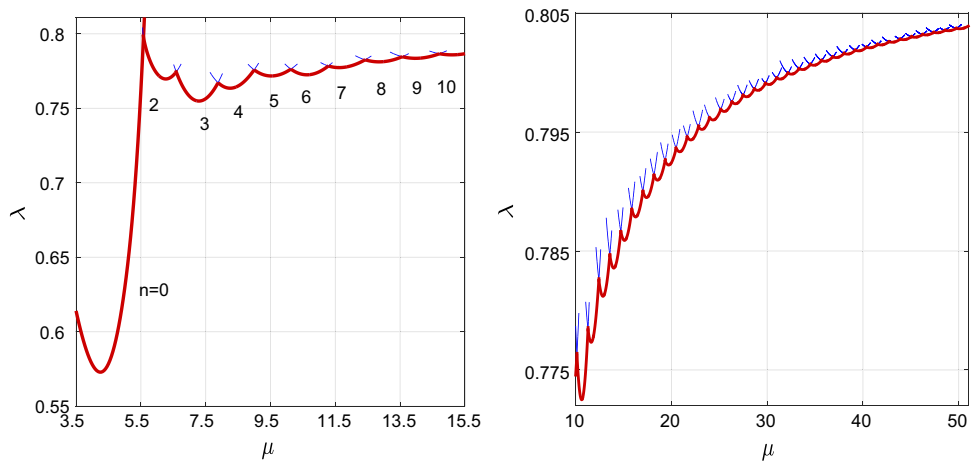

Fig. 4. Details of the neutral stability envelope for the externally pressurised problem $\mathcal{C}_{\text{sph}}^{(-)}$. In the left panel are shown parts of the curves corresponding to $m=0,2,3,4,\ldots,10$ while in the right panel are the curves taken to larger values $m\in[6,40]$.

the usual linearisation routine. The result is a set of similar equations in $\widehat{w}$ and $\widehat{F}$, which are not recorded here in the interest of brevity (they can be found in a number of places e.g., [11,12,15]). Solutions of these linearised bifurcation equations are then sought as functions with separable variables so that

$$\widehat{w}(\rho,\theta)=W(\rho)\cos(m\theta),\quad \widehat{F}(\rho,\theta)=\Psi(\rho)\cos(m\theta), \tag{3.1}$$

with the integer $m\geq 0$. The solution of the bifurcation equations will yield a value of the dimensionless applied pressure λ for each m; our goal then is to determine that choice of m which corresponds to the smallest λ. If we define $\boldsymbol{U}:=[W,\ \Psi]^T$, the unknown amplitudes in (3.1) satisfy the vector ordinary differential equation

$$\frac{d^4\boldsymbol{U}}{d\rho^4}+\boldsymbol{A}_3\frac{d^3\boldsymbol{U}}{d\rho^3}+\boldsymbol{A}_2\frac{d^2\boldsymbol{U}}{d\rho^2}+\boldsymbol{A}_1\frac{d\boldsymbol{U}}{d\rho}+\boldsymbol{A}_0\boldsymbol{U}=\boldsymbol{0}, \tag{3.2}$$

where

$$\boldsymbol{A}_3:=\begin{bmatrix}\mathcal{A}_{11}^{(3)} & 0\\ 0 & \mathcal{A}_{22}^{(3)}\end{bmatrix},\quad \boldsymbol{A}_2:=\begin{bmatrix}\mathcal{A}_{11}^{(2)} & \mathcal{A}_{12}^{(2)}\\ \mathcal{A}_{21}^{(2)} & \mathcal{A}_{22}^{(2)}\end{bmatrix},$$
$$\boldsymbol{A}_1:=\begin{bmatrix}\mathcal{A}_{11}^{(1)} & \mathcal{A}_{12}^{(1)}\\ \mathcal{A}_{21}^{(1)} & \mathcal{A}_{22}^{(1)}\end{bmatrix}\quad\text{and}\quad \boldsymbol{A}_0:=\begin{bmatrix}\mathcal{A}_{11}^{(0)} & \mathcal{A}_{12}^{(0)}\\ \mathcal{A}_{21}^{(0)} & \mathcal{A}_{22}^{(0)}\end{bmatrix}.$$

The definitions of the coefficients $\mathcal{A}_{ij}^{(k)}$ are listed in Appendix A, while the structure of the $\boldsymbol{A}_j$ matrices implies that this system is of order eight and consists of two coupled equations in W and Ψ. In the case of problem $\mathcal{C}_{\text{sph}}^{(-)}$ the system (3.2) needs to be solved subject to appropriate regularity conditions at the centre of the spherical cap, and fully clamped boundary conditions along its rim. In dimensionless form these boundary conditions are equivalent to

$$W(0)=\frac{dW}{d\rho}(0)=\Psi(0)=\frac{d\Psi}{d\rho}(0)=0, \tag{3.3a}$$

$$W(1)=\frac{dW}{d\rho}(1)=0, \tag{3.3b}$$

together with

$$\frac{d^2\Psi}{d\rho^2}(1)-\nu\left[\frac{d\Psi}{d\rho}(1)-m^2\Psi(1)\right]=0, \tag{3.4a}$$

$$\frac{d^3\Psi}{d\rho^3}(1)-\left[(1-\nu)+m^2(2+\nu)\right]\frac{d\Psi}{d\rho}(1)+3m^2\Psi(1)=0. \tag{3.4b}$$

The two conditions comprising (3.4) express the fact that the edge of the shell is restricted to move in the r- and θ-directions respectively. On the other hand, for the problem $\mathcal{C}_{\text{sph}}^{(+)}$ the conditions (3.4) should be replaced by $N_{rr}=N_{r\theta}=0$, whereby with the help of (2.2) it follows that

$$\Psi(1)=\frac{d\Psi}{d\rho}(1)=0. \tag{3.5}$$

Of course, the matrices $\boldsymbol{A}_j$ $(j=0,1,\ldots,3)$ depend on the mode number m, the aspect ratio μ and the dimensionless pressure (λ), as well as the pre-buckling fields (2.5). Therefore the system (3.2) subject to (3.3) and either (3.4) or (3.5) constitutes an eigenproblem for $\lambda=\lambda(m;\mu)$ where, for a given μ, the mode m corresponding to the smallest λ is the one of most significance.

In order to understand how the solutions of the eigenproblem might behave, it is helpful to first look at some direct numerical simulations and the corresponding structure of the neutral stability envelopes. As already mentioned our main interest is in the critical eigenvalues and mode numbers, which will be denoted by $\lambda_C>0$ and $m_C\in\mathbb{R}$, respectively; in words, m_C is the positive integer for which λ is least, with $\lambda_C=\lambda(m_C)$.

4. Numerical simulations

The numerical integration of $\mathcal{C}_{\text{sph}}^{(\pm)}$ was accomplished by writing the differential equation (2.4) and (3.2), together with the accompanying boundary conditions, as a first-order system in twelve variables; all the calculations described below were carried out by using standard routines in MATLAB, and were checked against existing results available in the literature (e.g., [12,13,16]).

Figs. 4 and 5 illustrate the typical neutral stability envelopes for the externally and internally pressurised spherical caps. For the $\mathcal{C}_{\text{sph}}^{(-)}$ problem it is known that the eigenproblem (3.2)–(3.4) has no solution for $\mu\lesssim 3.5$ (e.g., see [11,12,26]). Furthermore, as can be seen from Fig. 4(a), the axisymmetric mode ($m=0$) is the critical one only over a limited μ–range up to about $\mu\approx 5.5$. Once μ is slightly larger the next critical mode has $m=2$, and then as μ grows further so m_C increases in tandem. For each m the corresponding $\lambda(\mu)$ curve exists over a large range of μ, but in an attempt to avoid an unnecessarily intricate diagram, on Fig. 4(a) we only show for each m small sections of curve around those values for μ for which that m is the critical mode number m_C. The assemblage of the critical portions of the curves $\lambda=\lambda(\mu)$ together form the neutral stability envelope (shown in 'thick line'). More details on the behaviour of this envelope as μ grows appear in Fig. 4(b), where we consider a more extensive set of consecutive mode numbers with $6\leq m\leq 40$.

It is observed that λ_C increases with μ, albeit rather slowly. Additional experiments suggested that as $\mu \to \infty$ so the associated λ_C continues to grow monotonically and probably approaches a limiting constant; confirmation of this behaviour, backed up by an asymptotic study of the problem, follows from rational boundary-layer arguments that will be discussed in Section 5.2. Moreover, according to Fig. 4 it would appear that the critical $m = m_C$ also grows with μ.

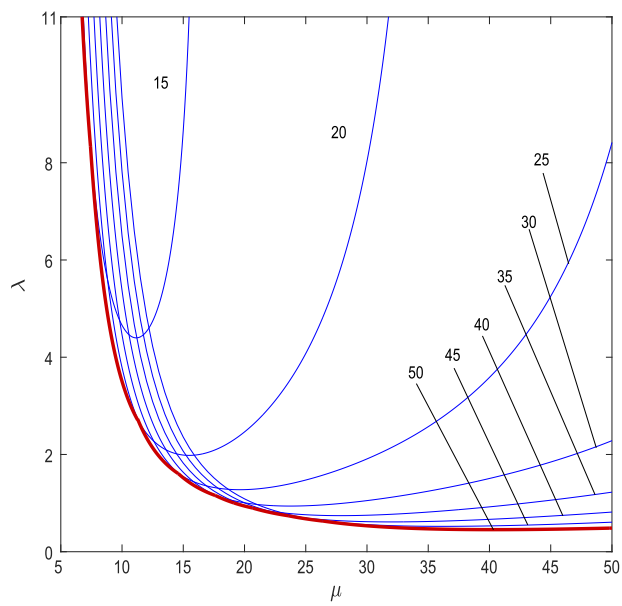

Fig. 5. A sample of curves $\lambda = \lambda(\mu)$ for $\mathcal{C}^{(+)}_{\text{sph}}$ that together form the neutral stability envelope. The various values of m used are as indicated.

The neutral stability envelope for the internally pressurised shell is more difficult to visualise owing to the somewhat awkward topology of the response curves $\lambda = \lambda(\mu)$. Nevertheless, we include a sample of the curves in Fig. 5, and it seems that the large-μ behaviour of the envelope contrasts to that for the $\mathcal{C}^{(-)}_{\text{sph}}$ case. We speculate that $\lambda_C \to 0$ as $\mu \to \infty$, a prediction that will be corroborated in due course by our analytical work, although the identity of the critical $m = m_C$ appears to grow with μ as it did in the $\mathcal{C}^{(-)}_{\text{sph}}$ problem.

The structures of the prebuckling fields and the wrinkling eigenmodes as $\mu \to \infty$ provide some of the main clues in isolating analytical approximations for λ_C and m_C. Solutions of both (2.4) and (3.2) experience strong localisation near the rim of the shell as μ becomes very large. As wrinkling in the circumferential direction is associated with sufficiently strong compressive pre-buckling hoop stresses, our eigenmode plots also include information on both $\mathring{N}_{rr}$ and $\mathring{N}_{\theta\theta}$, whose expressions are easily deduced from (2.2) since

$$\mathring{N}_{rr} \equiv \mathring{N}_{rr}(\rho) = \Phi/\rho \quad \text{and} \quad \mathring{N}_{\theta\theta} \equiv \mathring{N}_{\theta\theta}(\rho) = d\Phi/d\rho. \tag{4.1}$$

(The superimposed 'over-circle' on these fields is used here in order to differentiate them from their counterparts associated with the bifurcated solution.)

Fig. 6 illustrates the localisation of the transverse displacements W in (3.2) subject to (3.3) (or (3.4) for $\mathcal{C}^{(+)}_{\text{sph}}$) with the basic state given by (2.4) and (2.6) (or (2.7) for $\mathcal{C}^{(+)}_{\text{sph}}$). Four values of μ are shown; the eigenmode when $\mu = 6$ appears to extend over the entire cap $0 < \rho < 1$, a behaviour that can be ascribed to the fact that $\mathring{N}_{\theta\theta}$ is compressive over the whole range. Both the meridional

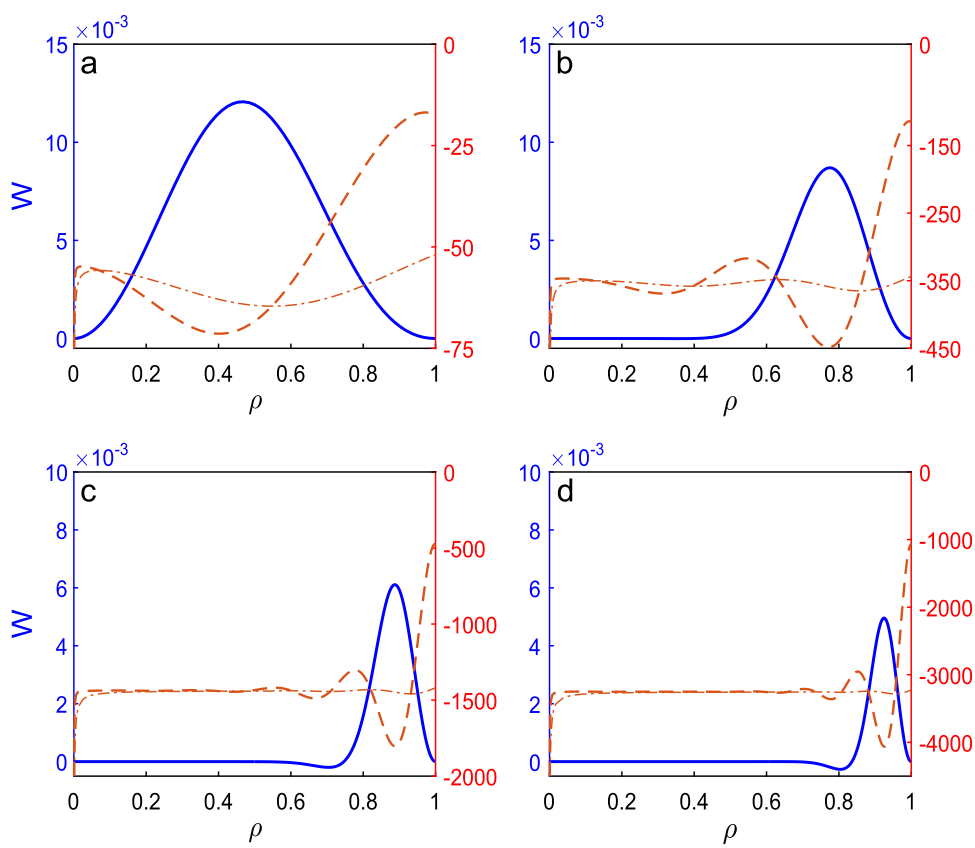

Fig. 6. The localisation of the eigenmodes for the externally pressurised spherical cap problem $\mathcal{C}^{(-)}_{\text{sph}}$ for values of μ:=(a) 6, (b) 15, (c) 30 and (d) 45. In each plot the solid curve shows the transverse displacement amplitude $W = W(\rho)$ in (3.1), with the corresponding vertical axis appearing on the left and the displacement normalised so that $\int_0^1 \Psi^2(\rho)\, d\rho = 0.1$. Superimposed on each diagram are the corresponding prebuckling membrane stresses (4.1): $N^{\backslash -\hat{}}{}_{rr}$ (shown by the dash-dot line) and $N^{\backslash -\hat{}}{}_{\theta\theta}$ (the dash line); the vertical axis for these fields appears on the right.

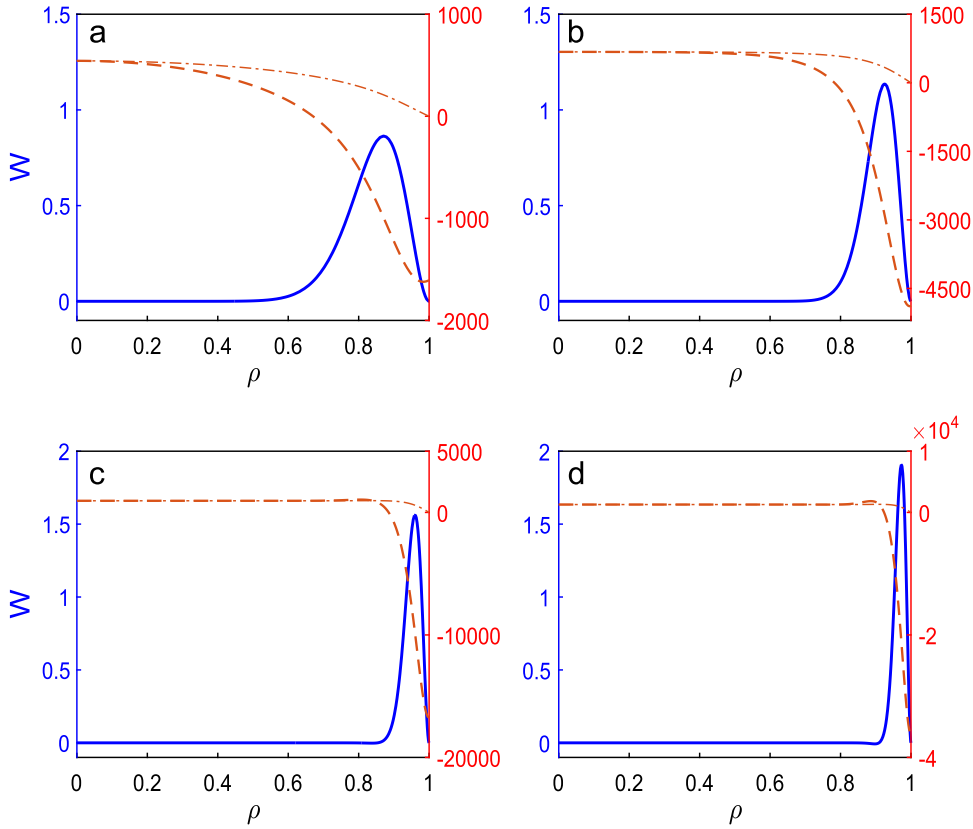

Fig. 7. Same as per Fig. 6, but for the internally pressurised spherical cap case $\mathcal{C}^{(+)}_{\text{sph}}$. The values of μ used are (a) 5, (b) 15, (c) 30 and (d) 45.

and hoop stresses remain compressive as μ increases, but the absolute minimum of the latter shifts towards the edge of the shell ($\rho=1$) and hence forces the eigenmodes to localise around this edge.

The localisation of the eigenmodes for the $\mathcal{C}^{(+)}_{\text{sph}}$ problem can be observed in Fig. 7, although the route to localisation seems to differ from that operative in $\mathcal{C}^{(-)}_{\text{sph}}$. It can be seen that $\mathring{N}_{rr}(\rho)>0$ for all $0<\rho<1$, irrespective of the size of μ. However, there exists a critical value of the radial coordinate, $\rho_0\equiv\rho_0(\mu)$ (say), at which $\dot{N}_{\theta\theta}(\rho)$ changes sign. Even for smallish $\mu=5$ it turns out that $\rho_0\simeq 0.65$ and the W – eigenmode is already fairly well localised – see, for example, Fig. 7(a). For progressively larger values of μ, the situations shown in the other three windows of Fig. 7 suggest that $\lim_{\mu\to\infty}\rho_0(\mu)=1$.

5. The large-μ limit for $\mathcal{C}^{(-)}_{\text{sph}}$

In the remainder of the paper we aim to provide rational asymptotic arguments that explain the structure of the bifurcated solutions in the limit $\mu\to\infty$; a closely related goal is to investigate the possibility of obtaining asymptotic approximations for the critical loads corresponding to the non-dimensional parameter λ defined in (2.3). Our analysis is informed by the numerical results presented above: it is clear from those results that both the basic state and the bifurcated solutions appear to be confined near the shell edge ($\rho=1$) and experience progressive localisation as $\mu\to\infty$. This feature makes it possible to obtain an analytical approximation for the non-linear basic state (2.4) and, implicitly, for the coefficients of the bifurcation equation (3.2), which in turn makes the latter problem amenable to an asymptotic analysis.

5.1. The structure of the base state for $\mu\gg 1$

A crucial observation in what follows is that the functions $\Phi(\rho)\equiv-2\mu^2\lambda\rho$ and $\Theta(\rho)\equiv 0$ satisfy Eqs. (2.4) identically and, moreover, they fulfil the left-hand boundary conditions in (2.6) at $\rho=0$. They however do not satisfy the requirements at $\rho=1$, implying that some kind of boundary-layer structure must be present at the rim. Furthermore, our numerical work has suggested that

$$\lambda=\lambda_0+\lambda_1\mu^{-1}+\lambda_2\mu^{-2}+\cdots, \tag{5.1}$$

with the likelihood that $\lambda_0<1$. The upshot is that the deformation over most of the spherical cap (what we shall call the *outer solution*, valid for $|\rho-1|=\mathcal{O}(1)$) will have the form

$$\Theta_{\text{out}}\equiv 0 \quad\text{and}\quad \Phi_{\text{out}}=-2\rho\mu^2(\lambda_0+\lambda_1\mu^{-1}+\lambda_2\mu^{-2}+\cdots). \tag{5.2}$$

Routine scaling arguments of the boundary-layer type transition region attached to $\rho=1$ show that the spatial extent of this zone is $\mathcal{O}(\mu^{-1})$, so we introduce the local variable $X=\mathcal{O}(1)$ defined by

$$\rho=1-\mu^{-1}X, \tag{5.3}$$

and then expand the base state according to

$$\begin{bmatrix}\Theta_{\text{inn}}\\ \Phi_{\text{inn}}\end{bmatrix}=\begin{bmatrix}0\\ -2\mu^2\lambda_0\end{bmatrix}+\mu\begin{bmatrix}\widehat{\theta}_0(X)\\ \widehat{\phi}_0(X)\end{bmatrix}+\begin{bmatrix}\widehat{\theta}_1(X)\\ \widehat{\phi}_1(X)\end{bmatrix}+\cdots. \tag{5.4}$$

These inner solutions need to be matched to the outer forms (5.2) as $X\to\infty$ and the rim conditions (2.6) require that

$$\widehat{\phi}_0'(0)=2\lambda_0\nu,\quad \widehat{\phi}_1'(0)=-\nu\widehat{\phi}_0(0),\quad \widehat{\theta}_k(0)=0\quad (k=1,2,\ldots), \tag{5.5}$$

where a dash denotes differentiation with respect to X.

The zeroth-order equations for the basic state inner problem follow easily by using the ansatz (5.4) in the axisymmetric shell

equations (2.4) so

$$\frac{d^2\widehat{\theta}_0}{dX^2}+2\lambda_0\widehat{\theta}_0+\widehat{\phi}_0=-2\lambda_1+2\lambda_0 X \quad \text{and} \quad \frac{d^2\widehat{\phi}_0}{dX^2}=\widehat{\theta}_0. \tag{5.6}$$

On putting

$$\alpha^2 := \frac{1}{2}(1-\lambda_0), \quad \beta^2 := \frac{1}{2}(1+\lambda_0)$$

and making use of the conditions (5.5) together with the requirement that $\widehat{\theta}_0 \to 0$ as $X \to \infty$, it follows quickly that

$$\widehat{\theta}_0(X) = A\exp(-\alpha X)\sin(\beta X), \tag{5.7}$$

for some arbitrary constant $A \in \mathbb{R}$, and then

$$\widehat{\phi}_0(X) = -2\lambda_1+2\lambda_0 X+A\left[2\alpha\beta\cos(\beta X)-\lambda_0\sin(\beta X)\right]\exp(-\alpha X). \tag{5.8}$$

The constant A that appears in the expressions of these functions is simply found by imposing the requirement on $\widehat{\phi}_0$ at $X=0$ – see the first condition in (5.5); thus, $A = 2\lambda_0(1-\nu)/\beta$, which completes the full solution of the zeroth-order problem.

Further algebra leads to the governing equations for the first-order corrections in the inner solution expansion (5.4). Written in matrix form these can be cast as

$$\begin{bmatrix} \frac{d^2}{dX^2}+2\lambda_0 & 1 \\ -1 & \frac{d^2}{dX^2} \end{bmatrix} \begin{bmatrix} \widehat{\theta}_1 \\ \widehat{\phi}_1 \end{bmatrix} = \begin{bmatrix} \Gamma_{11}^{(-)} \\ \Gamma_{12}^{(-)} \end{bmatrix}, \tag{5.9}$$

where

$$\Gamma_{11}^{(-)} := \widehat{\theta}_0{}' - 2\lambda_2 + 2\lambda_1 X - 2\lambda_0 X\widehat{\theta}_0 + \widehat{\theta}_0\widehat{\phi}_0 \quad \text{and}$$
$$\Gamma_{12}^{(-)} := \widehat{\phi}_0{}' - \frac{1}{2}\widehat{\theta}_0{}^2 - 2\lambda_0.$$

The solution of this rather complicated system can be derived by standard means, although the algebra rapidly becomes rather unwieldy. The relevant solution is best written as

$$\begin{bmatrix} \widehat{\theta}_1(X) \\ \widehat{\phi}_1{}'(X) \end{bmatrix} = \begin{bmatrix} 0 \\ 2\lambda_1 \end{bmatrix} + \exp(-\alpha X)\begin{bmatrix} S_1^{(-)} & S_2^{(-)} \\ T_1^{(-)} & T_2^{(-)} \end{bmatrix}\begin{bmatrix} \cos(\beta X) \\ \sin(\beta X) \end{bmatrix}$$
$$+\exp(-2\alpha X)\left\{\begin{bmatrix} S_3^{(-)} & S_4^{(-)} \\ T_3^{(-)} & T_4^{(-)} \end{bmatrix}\begin{bmatrix} \cos(2\beta X) \\ \sin(2\beta X) \end{bmatrix} + \begin{bmatrix} S_5^{(-)} \\ T_5^{(-)} \end{bmatrix}\right\}, \tag{5.10}$$

where the various coefficients are defined by

$$S_2^{(-)} := \frac{1}{2}AX\left(1+\frac{\lambda_1}{2\alpha}\right)+\frac{\lambda_1(\lambda_0+2)A}{2\lambda_0(\lambda_0+1)}+\alpha(1+2\nu)A$$
$$+\frac{(\lambda_0^2-1)(4\lambda_0^2+3\lambda_0+5)A^2}{2\alpha\beta(16\lambda_0^2-25)},$$
$$S_1^{(-)} := \frac{\lambda_1 AX}{4\beta}+\frac{(\lambda_0^2-1)A^2}{4\lambda_0+5}, \quad S_3^{(-)} := \frac{A^2(24\lambda_0^2-15)}{4(16\lambda_0^2-25)},$$
$$S_4^{(-)} := \frac{12\lambda_0\alpha\beta A^2}{16\lambda_0^2-25}, \quad S_5^{(-)} := -\frac{(2\lambda_0-1)^2A^2}{4(4\lambda_0-5)},$$

and

$$T_1^{(-)} := -\beta A\left[\frac{1}{4}\left(\frac{\lambda_1}{\alpha\beta^2}+2\right)X+\left(\frac{\lambda_1}{\lambda_0}+2\alpha\nu\right)\right]-\frac{6\lambda_0(\lambda_0^2-1)A^2}{\alpha(16\lambda_0^2-25)},$$
$$T_2^{(-)} := -\frac{1}{2}\alpha AX-\frac{1}{2}A\left[2\nu(1-\lambda_0)+1\right]$$
$$+\frac{\lambda_1(\lambda_0^2+\lambda_0-1)A}{4\lambda_0\alpha\beta^2}-\frac{(\lambda_0^2-1)(2\lambda_0+5)A^2}{\beta(16\lambda_0^2-25)},$$
$$T_3^{(-)} := -\frac{\alpha A^2(8\lambda_0-5)(\lambda_0+1)}{16\lambda_0^2-25}, \quad T_4^{(-)} := \frac{\beta A^2(8\lambda_0+5)(\lambda_0-1)}{16\lambda_0^2-25},$$
$$T_5^{(-)} := \frac{A^2(\lambda_0^2-1)}{2\alpha(4\lambda_0-5)}.$$

We remark that these forms have been verified by symbolic means; notice also that we record the expression for $\widehat{\phi}_1{}'$ rather than $\widehat{\phi}_1$ itself. Of course, the latter could be deduced following one integration but it will be seen in due course that it is the former expression that is really of eventual interest.

5.2. The bifurcated solution

We now turn our attention to the actual bifurcation problem. In previous works we have found that the disturbance structure can be somewhat intricate as it involves both the salient length scales present in the pre-buckle state and the relevant sizes of the radial and azimuthal derivatives of the perturbation. Given that the analysis above suggests that the base state develops just two principal forms, one for $\rho = \mathcal{O}(1) < 1$ and one at the rim where $1-\rho = \mathcal{O}(\mu^{-1})$, in turns out that on this rim scaling the highest order derivatives in (3.2) balance the azimuthal terms if $m = \mathcal{O}(\mu)$ and also incorporates variation in the pre-buckle state simultaneously. The implication is that the bifurcation is described by a very simple structure in terms of radial variables which consists only of the region where $1-\rho = \mathcal{O}(\mu^{-1})$. This means that the mathematical details are relatively easy because delicate balances do not arise, but the price that has to be paid is that the asymptotic equations are likely to only be marginally simpler than the original system (3.2) and necessitate numerical solution. Typically, if an asymptotic problem requires a sophisticated argument to establish the key structure, though the underlying reasons for the structure are often somewhat opaque, considerable analytic progress can be made (cf. [18,19]).

Given these observations, the structure of Eqs. (3.2) prompts us to look for a solution of the form

$$\begin{bmatrix} W \\ \Psi \end{bmatrix} = \begin{bmatrix} W_0(X) \\ \Psi_0(X) \end{bmatrix} + \mu^{-1}\begin{bmatrix} W_1(X) \\ \Psi_1(X) \end{bmatrix} + \cdots, \tag{5.11}$$

where the unknown functions W_k, Ψ_k $(k = 0, 1, \ldots)$ are to be found in the usual way by substituting (5.11) in (3.2) and comparing coefficients of like powers of μ.

At leading order we uncover the homogeneous system

$$\begin{bmatrix} \frac{d^4}{dX^4}+b_1^{(-)}\frac{d^2}{dX^2}+b_2^{(-)} & -\frac{d^2}{dX^2}+b_3^{(-)} \\ \frac{d^2}{dX^2}+b_4^{(-)} & \frac{d^4}{dX^4}+b_5^{(-)}\frac{d^2}{dX^2}+b_6^{(-)} \end{bmatrix}\begin{bmatrix} W_0 \\ \Psi_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{5.12}$$

where

$$b_1^{(-)} := -2(M^2-\lambda_0), \quad b_4^{(-)} := -M^2\left(1+\frac{d\widehat{\theta}_0}{dX}\right),$$
$$b_2^{(-)} := M^2\left(M^2-\frac{d\widehat{\phi}_0}{dX}\right), \quad b_5^{(-)} := -2M^2,$$
$$b_3^{(-)} := M^2\left(1+\frac{d\widehat{\theta}_0}{dX}\right), \quad b_6^{(-)} := M^4$$

and the mode number has been scaled by writing $m = \mu M$. The solutions of the eighth-order system (5.12) need to decay exponentially as they leave the boundary layer (i.e., as $X \to \infty$), while the boundary conditions (3.3) place the additional constraints

$$W_0 = W_0' = 0 \quad \text{at } X = 0, \tag{5.13a}$$

$$\Psi_0'' + \nu M^2\Psi_0 = 0 \quad \text{and} \quad \Psi_0''' - (2+\nu)M^2\Psi_0' = 0 \quad \text{at } X = 0. \tag{5.13b}$$

We note that (5.12) together with the boundary conditions and the aforementioned decay requirements constitutes an eigenproblem

Table 1
The coefficients in the formulae (5.20) for various values of ν.

ν	0.1	0.15	0.20	0.25	0.30	0.35	0.40	0.45
λ_{0C}	0.7689	0.7771	0.7858	0.7948	0.8044	0.8144	0.8249	0.8360
λ_{1C}	−0.3524	−0.3590	−0.3652	−0.3237	−0.3285	−0.3324	−0.2870	−0.2886
M_C	0.84	0.84	0.84	0.85	0.85	0.85	0.86	0.86

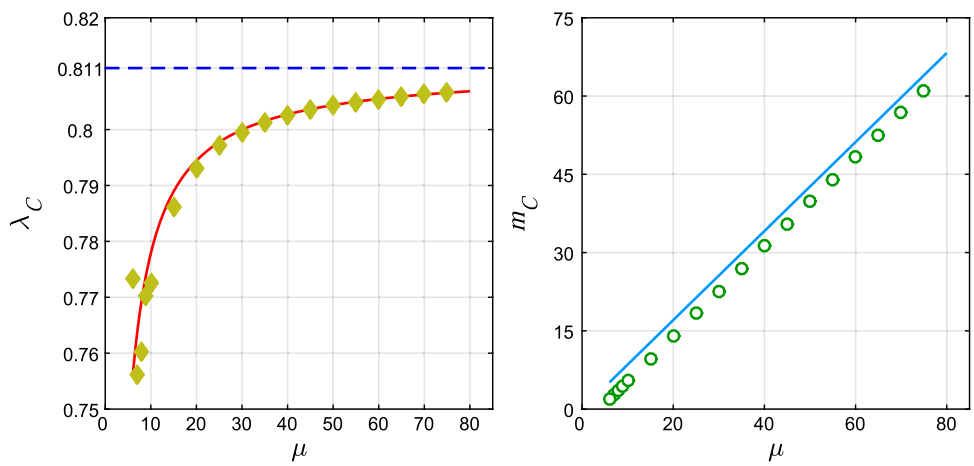

Fig. 8. Comparisons between asymptotic predictions and direct numerical simulations of the boundary-value problem $C_{\rm sph}^{(-)}$ for $\nu=1/3$ ($\lambda_{0C}\simeq 0.811$, $\lambda_{1C}\simeq -0.3312$ and $M_C\simeq 0.853$). The asymptotic formula for the critical wrinkling load (5.20) is shown at various values of μ between 6 and 75 in the left window, while the critical mode number m_C is indicated in the right window. The dotted line on the left-hand plot is the one-term asymptote $\lambda_{0C}\approx 0.811$.

in which the eigenvalue λ_0 depends on the parameter $M\in\mathbb{R}_+$. This problem is eighth-order, as is the full bifurcation system (3.2), but many of the coefficients are a little simpler and of course the effective basic state quantities $\widehat{\theta}_0$ and $\widehat{\phi}_0$ are already known. Nevertheless, this system still requires a computational solution and, as we are ultimately mainly interested in the critical wrinkling pressure, our aim is to minimise $\lambda_0=\lambda_0(M)$. Standard numerical work readily gives that, for $\nu=1/3$ (the value adopted in the numerical simulations discussed in Section 4) the minimum value of λ_0 occurs when $M=M_C\simeq 0.853$, and then $\lambda_0=\lambda_{0C}\simeq 0.811$. These critical values are functions of ν but, as can be seen from the data in Table 1, this dependence is relatively weak and, in particular, the size of M_C is almost independent of ν.

Huang [12] recognised the asymptotic structure of the bifurcation problem $C_{\rm sph}^{(-)}$, although his equations were written over $[0,\mu]$ and the asymptotic parameter appeared in the boundary conditions; also, only an ad hoc leading-order analysis was attempted (while the thickness of the boundary layers remained unclear). He found that $\lambda_{0C}\simeq 0.864$ and $M_C\simeq 0.817$. The discrepancy between these values and those calculated above can be traced directly to his over-simplification that $\widehat{\phi}_0(X)\simeq 2\lambda_0 X$ (cf. (5.8)) which changes slightly some of the coefficients appearing in system (5.12), and hence leads to the error in his results.

Although the approximation $\lambda_C\simeq\lambda_{0C}$ is of use in verifying that the critical pressure asymptotes to a constant value $\lambda_{0C}\approx 0.811$, which is certainly consistent with results shown in Fig. 4(b), it cannot capture the general shape of the critical envelope. This can perhaps be established by taking the expansion for λ to further orders. The substitution of (5.11) in (3.2) leads to the conclusion that

$$\begin{bmatrix} \frac{d^4}{dX^4}+b_1^{(-)}\frac{d^2}{dX^2}+b_2^{(-)} & -\frac{d^2}{dX^2}+b_3^{(-)} \\ \frac{d^2}{dX^2}+b_4^{(-)} & \frac{d^4}{dX^4}+b_5^{(-)}\frac{d^2}{dX^2}+b_6^{(-)} \end{bmatrix}\begin{bmatrix} W_1 \\ \Psi_1 \end{bmatrix}=\begin{bmatrix} \Delta_{11}^{(-)} \\ \Delta_{12}^{(-)} \end{bmatrix}, \tag{5.14}$$

where

$$\begin{aligned}\Delta_{11}^{(-)}:=&\,2W_0'''+(4M^2X-2\lambda_0X+\widehat{\phi}_0)W_0''+(\widehat{\phi}_0'+2M^2)W_0' \\ &-(4M^4X-2M^2X\widehat{\phi}_0'-M^2\widehat{\phi}_1')W_0-\widehat{\theta}_0\Psi_0''-(1+\widehat{\theta}_1')\Psi_0' \\ &-M^2\left[2X(1+\widehat{\theta}_0')+\widehat{\theta}_1'\right]\Psi_0,\end{aligned} \tag{5.15}$$

and

$$\begin{aligned}\Delta_{12}^{(-)}:=&\,2\Psi_0'''+\widehat{\theta}_0W_0''+4M^2X\Psi_0''+(1+\widehat{\theta}_0')W_0'+2M^2\Psi_0' \\ &+M^2\left[2X(1+\widehat{\theta}_0')+\widehat{\theta}_1'\right]W_0-4M^4X\Psi_0.\end{aligned} \tag{5.16}$$

At this order the boundary conditions for (5.14) become

$$W_1=W_1'=0 \quad \text{at } X=0, \tag{5.17a}$$

$$\Psi_1''+\nu M^2\Psi_1=-\nu\Psi_0' \quad \text{at } X=0, \tag{5.17b}$$

in addition to the requirement of exponential decay as $X\to\infty$.

Since the equations for (W_1,Ψ_1) are forced versions of the leading order equations for (W_0,Ψ_0), a solution of (5.14) subject to (5.17) is only possible if a certain compatibility condition is satisfied. It turns out that the forced problem is essentially self-adjoint which means that the necessary solvability condition can be deduced by multiplying the two equations comprising (5.14) by W_0 and Ψ_0 respectively. If the two results are subtracted and then integrated over $[0,\infty)$, repeated integration by parts and the knowledge that (W_0,Ψ_0) satisfy (5.12) leads to the conclusion that a solution of the problem for W_1 and Ψ_1 is only feasible if

$$\int_0^\infty\left[\Delta_{11}^{(-)}W_0-\Delta_{12}^{(-)}\Psi_0\right]dX=3M^2\Psi_{00}^2+\nu(\Psi_{00}')^2, \tag{5.18}$$

where Ψ_{00} and Ψ_{00}' are the values of Ψ_0 and its derivative at $X=0$. We also note that the basic state fields $\widehat{\phi}_0$, $\widehat{\phi}_1'$ and $\widehat{\theta}_1$ all depend linearly on λ_1, so we write

$$\widehat{\theta}_1=\lambda_1\theta_{11}+\theta_{10},\quad \widehat{\phi}_1'=\lambda_1\phi_{11}'+\phi_{10}',\quad \widehat{\phi}_0=-2\lambda_1+\phi_{00},$$

where all the new functions introduced on the right-hand sides of these equations can be inferred directly from (5.8) and (5.10). If the

definitions of $\Delta_{11}^{(-)}$ and $\Delta_{12}^{(-)}$ are used in (5.18), this criterion can be reduced to the form

$$\lambda_1\left[\int_0^\infty \mathcal{H}_1^{(-)}(X)\,dX\right]+\int_0^\infty \mathcal{H}_2^{(-)}(X)\,dX=(1+\nu)\left[2M^2\Psi_{00}^2+(\Psi_{00}')^2\right], \tag{5.19}$$

where

$$\mathcal{H}_1^{(-)}(X):=M^2\phi_{11}'W_0^2+2(W_0')^2+2M^2\theta_{11}(W_0\Psi_0)'$$

and

$$\begin{aligned}\mathcal{H}_2^{(-)}(X):=&\,4M^2X(W_0W_0''-\Psi_0\Psi_0'')-2\lambda_0XW_0W_0''-\phi_{00}(W_0')^2\\&-4XM^4(W_0^2-\Psi_0^2)+M^2\left(2X\widehat{\phi}_0'+\widehat{\phi}_{10}'\right)W_0^2\\&-4M^2X\left(1+\widehat{\theta}_0'\right)W_0\Psi_0+2\widehat{\theta}_0W_0'\Psi_0'+2M^2\theta_{10}(W_0\Psi_0)'.\end{aligned}$$

The value of λ_{1C} is found from (5.19) by setting $M\to M_C$, $\lambda_0\to\lambda_{0C}$, using the zeroth-order fields W_0, Ψ_0 associated with these critical values, and then performing the integrals numerically. The conclusion is that the final asymptotic results for the $\mathcal{C}_{\text{sph}}^{(-)}$ problem are

$$\lambda_C\simeq\lambda_{0C}+\lambda_{1C}\mu^{-1}+\mathcal{O}(\mu^{-2})\quad\text{and}\quad m_C\simeq M_C\mu+\mathcal{O}(1), \tag{5.20}$$

with the coefficients that appear as recorded in Table 1.

Some comparisons between the estimates predicted by (5.20) and direct numerical simulations of $\mathcal{C}_{\text{sph}}^{(-)}$ are illustrated in Fig. 8 where again we have used $\nu=1/3$. The markers in the left window indicate the form of the first equation of (5.20), the solid line was obtained by solving the full boundary-value problem for the externally pressurised spherical cap when $\mu\in[6,75]$. The dotted line denotes the constant λ_{0C} and is clear that the two-term result is in excellent accord with the direct numerical simulations even for values of μ as small as about 10. Indeed the relative error of the two-term asymptote is only 0.68% when $\mu=10$ and barely 0.02% once $\mu=40$. This is positive confirmation of the predictive power of our asymptotic work. The approximations for the number of wrinkles are included in the right window of Fig. 8, with the markers indicating the values computed from the simple result in (5.20) and the continuous line denoting the critical mode number obtained by integrating numerically $\mathcal{C}_{\text{sph}}^{(-)}$. The agreement now is less impressive but this is unsurprising when it is remembered that finding the critical values of λ and m is akin to locating the minimum of a locally parabolic curve. One would expect that the m-coordinate would be the one which contains the greater error; of course, if determined, one could deduce further terms in the expansion for m_C but the effort required is probably not commensurate with the improvement achieved. As it is, one can expect the preferred wrinkle pattern to contain a number of folds that increases linearly with μ; moreover, we have seen that the critical external pressure, which is arguably the quantity of greater physical significance, can be found with minimal error for values of μ as low as 10.

6. The large-μ limit for $\mathcal{C}_{\text{sph}}^{(+)}$

While the bifurcation mechanism for the internally pressurised spherical cap is fairly different from the case analysed in Section 5, the underlying mathematical structure turns out to be remarkably similar. Accordingly, we keep our description of the asymptotics of the $\mathcal{C}_{\text{sph}}^{(+)}$ problem to a bare minimum and only note the main differences from what has been described already.

As in Section 5.1, except for an $\mathcal{O}(\mu^{-1})$-depth zone near $\rho=1$, the basic state is described again by the outer solution $\Phi(\rho)=2\mu^2\lambda\rho$ and $\Theta(\rho)\equiv 0$. Understanding the boundary layer requires the introduction of the same rescaled coordinate $X=\mathcal{O}(1)$ as in (5.3), while numerical evidence suggests that (5.1) must now be replaced by

$$\lambda=\widetilde{\lambda}_0\mu^{-1}+\widetilde{\lambda}_1\mu^{-2}+\cdots, \tag{6.1}$$

since $\lambda\to 0$ as $\mu\to\infty$; the new coefficients $\widetilde{\lambda}_0$ and $\widetilde{\lambda}_1$ are determined below.

The basic state inner solution is expanded according to

$$\begin{bmatrix}\Theta_{\text{inn}}\\ \Phi_{\text{inn}}\end{bmatrix}=\mu\begin{bmatrix}\Theta_0(X)\\ \Phi_0(X)\end{bmatrix}+\begin{bmatrix}\Theta_1(X)\\ \Phi_1(X)\end{bmatrix}+\cdots. \tag{6.2}$$

The main difference between this and the externally pressurised problem is that since the eigenvalue λ is smaller, this tends to generate simpler equations. For instance the governing equations for Θ_0 and Φ_0 are

$$\frac{d^2\Theta_0}{dX^2}+\Phi_0=2\widetilde{\lambda}_0\quad\text{and}\quad\frac{d^2\Phi_0}{dX^2}=\Theta_0, \tag{6.3}$$

(cf. Eqs. (5.6) for the $\mathcal{C}_{\text{sph}}^{(-)}$ problem). The solution of these, which satisfies the rim conditions and matches to the outer solution, is

$$\begin{aligned}&\Theta_0(X)=-2\widetilde{\lambda}_0\exp(-X/\sqrt{2})\sin(X/\sqrt{2})\quad\text{and}\\&\Phi_0(X)=2\widetilde{\lambda}_0\left[1-\exp(-X/\sqrt{2})\cos(X/\sqrt{2})\right].\end{aligned} \tag{6.4}$$

The solution of the next-order problem may be expressed as

$$\begin{aligned}\begin{bmatrix}\Theta_1(X)\\ \Phi_1(X)\end{bmatrix}=&\begin{bmatrix}0\\ 2(\widetilde{\lambda}_1-\widetilde{\lambda}_0X)\end{bmatrix}+\exp(-X/\sqrt{2})\begin{bmatrix}S_1^{(+)} & S_2^{(+)}\\ T_1^{(+)} & T_2^{(+)}\end{bmatrix}\begin{bmatrix}\cos(X/\sqrt{2})\\ \sin(X/\sqrt{2})\end{bmatrix}\\&+\exp(-X\sqrt{2})\left\{\begin{bmatrix}S_3^{(+)} & 0\\ 0 & T_4^{(+)}\end{bmatrix}\begin{bmatrix}\cos(X\sqrt{2})\\ \sin(X\sqrt{2})\end{bmatrix}+\begin{bmatrix}S_5^{(+)}\\ T_5^{(+)}\end{bmatrix}\right\},\end{aligned} \tag{6.5}$$

where

$$S_1^{(+)}:=\widetilde{\lambda}_0^2\left(\frac{X}{\sqrt{2}}-\frac{4}{5}\right),\quad S_2^{(+)}:=2\left(\frac{\widetilde{\lambda}_0^2}{5}-\widetilde{\lambda}_1\right)+\widetilde{\lambda}_0X\left(\frac{\widetilde{\lambda}_0}{\sqrt{2}}-1\right),\quad S_3^{(+)}:=\frac{3\widetilde{\lambda}_0^2}{5},$$

$$S_5^{(+)}:=\frac{\widetilde{\lambda}_0^2}{5},\quad T_1^{(+)}:=S_2^{(+)},\quad T_2^{(+)}:=-\widetilde{\lambda}_0^2\left(\frac{X}{\sqrt{2}}+\frac{6}{5}\right),\quad T_4^{(+)}\equiv T_5^{(+)}:=-\frac{2\widetilde{\lambda}_0^2}{5}.$$

For the bifurcation equation (3.2) we shall use the familiar ansatz

$$\begin{bmatrix}W\\ \Psi\end{bmatrix}=\begin{bmatrix}\widetilde{W}_0(X)\\ \widetilde{\Psi}_0(X)\end{bmatrix}+\mu^{-1}\begin{bmatrix}\widetilde{W}_1(X)\\ \widetilde{\Psi}_1(X)\end{bmatrix}+\cdots, \tag{6.6}$$

where the leading-order terms are given by solving the homogeneous problem

$$\begin{bmatrix}\frac{d^4}{dX^4}+b_1^{(+)}\frac{d^2}{dX^2}+b_2^{(+)} & -\frac{d^2}{dX^2}+b_3^{(+)}\\ \frac{d^2}{dX^2}+b_4^{(+)} & \frac{d^4}{dX^4}+b_5^{(+)}\frac{d^2}{dX^2}+b_6^{(+)}\end{bmatrix}\begin{bmatrix}\widetilde{W}_0\\ \widetilde{\Psi}_0\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}, \tag{6.7}$$

with the coefficients

$$b_1^{(+)}:=-2M^2,\quad b_4^{(+)}:=-M^2\left(1+\frac{d\Theta_0}{dX}\right),$$

$$b_2^{(+)}:=M^2\left(M^2-\frac{d\Phi_0}{dX}\right),\quad b_5^{(+)}:=-2M^2,$$

$$b_3^{(+)}:=M^2\left(1+\frac{d\Theta_0}{dX}\right),\quad b_6^{(+)}:=M^4.$$

The mode number has again been scaled according to $m=\mu M$, with $M=\mathcal{O}(1)$. This eighth-order eigen-system needs to be solved subject to $\widetilde{W}_0$, $\widetilde{W}_0'$, $\widetilde{\Psi}$, $\widetilde{\Psi}_0'=0$ at $X=0$ and as $X\to\infty$. Our numerical routines developed for the previous calculations make it relatively straightforward to find the dependence of $\widetilde{\lambda}_0$ on $M>0$, and hence obtain the value $M=M_C$ for which $\widetilde{\lambda}_0$ is minimum, say, $\widetilde{\lambda}_{0C}:=\widetilde{\lambda}_0(M_C)$.

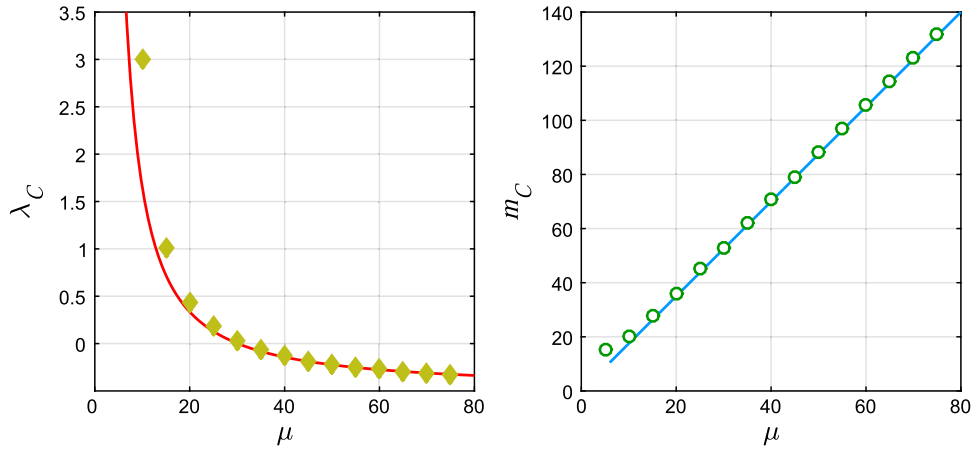

Fig. 9. Comparisons between the asymptotic results and direct numerical simulations for the problem $\mathcal{C}_{\rm sph}^{(+)}$. On the left, the critical wrinkling load λ_C given by (6.14) is shown as the continuous line, while the markers represent direct numerical simulations of the problem $\mathcal{C}_{\rm sph}^{(+)}$; here $\mu=5j$, with $j=1,2,\ldots,15$. A similar comparison for the critical mode number is included in the right-hand side window.

The next-order equations for $\widetilde{W}_1$ and $\widetilde{\Psi}_1$ can be used to derive a compatibility condition as in Section 5.2, which will then lead to an expression for $\widetilde{\lambda}_1$ in (6.1). Routine algebraic manipulations show that

$$\begin{bmatrix} \frac{d^4}{dX^4}+b_1^{(+)}\frac{d^2}{dX^2}+b_2^{(+)} & -\frac{d^2}{dX^2}+b_3^{(+)} \\ \frac{d^2}{dX^2}+b_4^{(+)} & \frac{d^4}{dX^4}+b_5^{(+)}\frac{d^2}{dX^2}+b_6^{(+)} \end{bmatrix} \begin{bmatrix} \widetilde{W}_1 \\ \widetilde{\Psi}_1 \end{bmatrix} = \begin{bmatrix} \Delta_{11}^{(+)} \\ \Delta_{12}^{(+)} \end{bmatrix}, \tag{6.8}$$

where

$$\begin{aligned} \Delta_{11}^{(+)} := {} & 2\widetilde{W}_0{}''' + (4M^2X+\Phi_0)\widetilde{W}_0{}'' + (\Phi_0'+2M^2)\widetilde{W}_0{}' \\ & -(4M^4X-2M^2X\Phi_0'-M^2\Phi_1')\widetilde{W}_0 - \Theta_0\widetilde{\Psi}_0{}'' - (1+\Theta_0')\widetilde{\Psi}_0{}' \\ & -M^2\left[2X(1+\Theta_0')+\Theta_1'\right]\widetilde{\Psi}_0 \end{aligned} \tag{6.9}$$

and

$$\begin{aligned} \Delta_{12}^{(+)} := {} & 2\widetilde{\Psi}_0{}''' + \Theta_0\widetilde{W}_0{}'' + 4M^2X\widetilde{\Psi}_0{}'' + (1+\Theta_0')\widetilde{W}_0{}' + 2M^2\widetilde{\Psi}_0{}' \\ & +M^2\left[2X(1+\Theta_0')+\Theta_0'\right]\widetilde{W}_0 - 4M^4X\widetilde{\Psi}_0. \end{aligned} \tag{6.10}$$

A solution of the system (6.8) is required, with $\widetilde{W}_1$ and $\widetilde{\Psi}_1$ and their first derivatives vanishing both on $X=0$ and as $X\to\infty$. This is possible only if

$$\int_0^\infty \left[\Delta_{11}^{(+)}\widetilde{W}_0 - \Delta_{12}^{(+)}\widetilde{\Psi}_0\right] dX = 0, \tag{6.11}$$

which reduces to

$$4M^2\widetilde{\lambda}_1\left[\int_0^\infty \mathcal{H}_1^{(+)}(X)\,dX\right] = \int_0^\infty \mathcal{H}_2^{(+)}(X)\,dX, \tag{6.12}$$

where

$$\begin{aligned} \mathcal{H}_1^{(+)}(X) := {} & \widetilde{W}_0\widetilde{W}_0{}'\left[1-\exp(-X/\sqrt{2})\cos(X/\sqrt{2})\right] \\ & +(\widetilde{W}_0\widetilde{\Psi}_0)'\exp(-X/\sqrt{2})\sin(X/\sqrt{2}) \end{aligned}$$

and

$$\begin{aligned} \mathcal{H}_2^{(+)}(X) := {} & 4M^2X(\widetilde{W}_0\widetilde{W}_0{}'' - \widetilde{\Psi}_0\widetilde{\Psi}_0{}'') + 2\Theta_0\widetilde{W}_0{}'\widetilde{\Psi}_0{}' - 4M^4X(\widetilde{W}_0^2-\widetilde{\Psi}_0^2) \\ & +2M^2X\widetilde{\Phi}_0{}'\widetilde{W}_0^2 - 2M\widetilde{W}_0\widetilde{W}_0{}'\Phi_{10} - 4M^2X(1+\Theta_0')\widetilde{W}_0\widetilde{\Psi}_0 \\ & +2M^2\Theta_{10}\left(\widetilde{W}_0\widetilde{\Psi}_0\right)' - \Phi_0\left(\widetilde{W}_0{}'\right)^2. \end{aligned} \tag{6.13}$$

We remark that the numerical problem for $\widetilde{W}_0$ and $\widetilde{\Psi}_0$ is independent of ν so that for this internally pressurised case we can compute the various eigenvalues once and for all. It follows that at critical conditions

$$\lambda_C \simeq 11.8753\mu^{-1} + 96.4626\mu^{-2} + \mathcal{O}(\mu^{-3}) \quad \text{and} \quad m_C \simeq 1.75\mu. \tag{6.14}$$

Some comparisons between (6.14) and direct numerical simulations of the full problem $\mathcal{C}_{\rm sph}^{(+)}$ are illustrated in Fig. 9. In terms of the critical dimensionless wrinkling pressure, the asymptotic expressions are very good for large values of μ (e.g., the relative error is less than 1% for $\mu=75$ and about 2.8% for $\mu=40$), but naturally they deteriorate in accuracy as μ decreases. Similar observations can be made in respect of the asymptotic formula for m_C which seems to compare excellently with the numerical results.

7. Concluding remarks

The edge-wrinkling experienced by a shallow spherical shell subjected to uniform pressure loads has been investigated here in the limit $\mu \gg 1$, where μ is the aspect-ratio quantity defined in (2.3a). It has been shown that both the externally and the internally pressurised problems have the same asymptotic structure despite the fact that the physical mechanisms responsible for the localisation of the wrinkling instability patterns are essentially different (as explained in Section 4). Our main results consist of two-term asymptotic predictions for the critical pressure λ_C and one-term approximations of the critical mode number m_C, and these are summarised in Eqs. (5.20) and (6.14).

In related recent work [18,19] that dealt with a uniformly stretched pressurised thin elastic plate, it was discovered that the asymptotic structure was controlled by the interaction between a novel membrane boundary-layer associated with the mixed nature of the hoop membrane stresses (partly compressive/tensile) and the usual bending layer generated by the presence of very little bending stiffness. Rather unexpectedly, our present analysis revealed simpler asymptotic phenomena that rely essentially only on bending effects in both the basic state (2.4) and the bifurcation equations (3.2).

One of the main differences between the current work and that reported in the aforementioned references is the lack of initial tension. This can be easily accommodated by modifying the second equation in (2.7a) to

$$\Phi(1) = \mathcal{F}, \quad \mathcal{F} \equiv \frac{\mu^3}{4}\left(\frac{a}{H}\right)^2\left(\frac{\mathcal{F}^*}{Eh}\right), \tag{7.1}$$

which corresponds to prescribing a given edge membrane force;

here $\mathcal{F}^*$ is the dimensional counterpart of $\mathcal{F}$ and the convention is that tensile loads correspond to $\mathcal{F} < 0$. Shilkrut [16] has presented some numerical results for this case in the context of our $\mathcal{C}_{\text{sph}}^{(+)}$ problem, and suggested that the localisation of the deformation patterns follows from increasing the absolute value of $\mathcal{F}$; since (7.1) taken at face value implies that $\mathcal{F} \propto \mu^3$ this conclusion seems natural. We have not pursued this case in detail because the scaling of both the basic state and the bifurcation equation adopted herein are not entirely optimal for this new problem. As indicated by the presence of P_{sph} in the definition of our λ – see (2.3b), the scalings in Section 2 were basically motivated by the analogy between the problem $\mathcal{C}_{\text{sph}}^{(-)}$ and the buckling of a complete spherical shell; however, the internally pressurised spherical cap subjected to tensile tractions along its rim is only remotely related to the latter problem. This modified situation and several other aspects are currently the object of an ongoing investigation, whose findings we hope to report in due course.

Appendix A. Coefficients of the bifurcation equations

The coefficients $A_{ij}^{(k)}$ that appear in (3.2) are given by

$\boldsymbol{A}_3$:

$$\mathcal{A}_{11}^{(3)} \equiv \mathcal{A}_{22}^{(3)} = \frac{2}{\rho},$$

$\boldsymbol{A}_2$:

$$\mathcal{A}_{11}^{(2)} := -\frac{1}{\rho}\left(\frac{2m^2+1}{\rho} + \Phi\right), \quad \mathcal{A}_{12}^{(2)} := -\mu^2 + \frac{\Theta}{\rho},$$

$$\mathcal{A}_{21}^{(2)} = -\mathcal{A}_{12}^{(2)}, \quad \mathcal{A}_{22}^{(2)} := -\frac{2m^2+1}{\rho^2},$$

$\boldsymbol{A}_1$:

$$\mathcal{A}_{11}^{(1)} := \frac{1}{\rho}\left(\frac{2m^2+1}{\rho^2} - \frac{d\Phi}{d\rho}\right), \quad \mathcal{A}_{12}^{(1)} := -\frac{1}{\rho}\left(\mu^2 - \frac{d\Theta}{d\rho}\right),$$

$$\mathcal{A}_{21}^{(1)} = -\mathcal{A}_{12}^{(1)}, \quad \mathcal{A}_{22}^{(1)} = \frac{2m^2+1}{\rho^3},$$

$\boldsymbol{A}_0$:

$$\mathcal{A}_{11}^{(0)} := \frac{m^2}{\rho^2}\left(\frac{m^2-4}{\rho^2} + \frac{d\Phi}{d\rho}\right), \quad \mathcal{A}_{12}^{(0)} := \frac{m^2}{\rho^2}\left(\mu^2 - \frac{d\Theta}{d\rho}\right),$$

$$\mathcal{A}_{21}^{(0)} = -\mathcal{A}_{12}^{(0)}, \quad \mathcal{A}_{22}^{(0)} := \frac{m^2(m^2-4)}{\rho^4}.$$

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